Cisco Virtual Internship 2025 - Supporting Document

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This document explains the network design and configuration created in Cisco Packet Tracer for the campus topology. It describes VLAN segmentation, router-on-a-stick configuration, DHCP services, wireless access, ACL-based security, and device hardening. The design follows the cybersecurity problem statement requirements from Cisco's VIP 2025 challenge.

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1. Network Topology

The campus network consists of three academic blocks, a library, student labs, a DMZ, and an open Wi-Fi zone (canteen).

- **Cisco 2911 Router:** Provides inter-VLAN routing and DHCP services.
- **Cisco 3650 Distribution Switch:** Connects to access switches.
- **Cisco 2960 Access Switches:** Provide connectivity for end devices.
- **WRT300N Wireless Router:** Provides Wi-Fi access for VLAN 60.
- **DMZ VLAN (70):** Hosts public-facing servers.

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2. VLANs & Subnets

VLAN ID	Zone	Subnet	Gateway
10	Admin	192.168.10.0/24	192.168.10.1
20	IT	192.168.20.0/24	192.168.20.1
30	Students	192.168.30.0/24	192.168.30.1
40	Library	192.168.40.0/24	192.168.40.1
50	Labs	192.168.50.0/24	192.168.50.1
60	Wi-Fi	192.168.60.0/24	192.168.60.1
70	DMZ	192.168.70.0/24	192.168.70.1

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3. Router Configuration (2911)

The router is configured as **router-on-a-stick** with subinterfaces for each VLAN.

```
interface g0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0

interface g0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0

! (similar for VLANs 30,40,50,60,70)
```

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4. DHCP Configuration

DHCP pools provide IPs to all VLANs. Gateway IPs are excluded.

```
ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1
ip dhcp excluded-address 192.168.30.1
ip dhcp excluded-address 192.168.40.1
ip dhcp excluded-address 192.168.50.1
ip dhcp excluded-address 192.168.60.1
ip dhcp excluded-address 192.168.70.1

! VLAN 10 - Admin
ip dhcp pool VLAN10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8

! VLAN 20 - IT
ip dhcp pool VLAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8

! VLAN 30 - Students
ip dhcp pool VLAN30
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8

! VLAN 40 - Library
ip dhcp pool VLAN40
 network 192.168.40.0 255.255.255.0
 default-router 192.168.40.1
 dns-server 8.8.8.8

! VLAN 50 - Labs
ip dhcp pool VLAN50
 network 192.168.50.0 255.255.255.0
 default-router 192.168.50.1
 dns-server 8.8.8.8

! VLAN 60 - Wi-Fi
```

```
ip dhcp pool VLAN60
 network 192.168.60.0 255.255.255.0
 default-router 192.168.60.1
 dns-server 8.8.8.8

! VLAN 70 - DMZ
ip dhcp pool VLAN70
 network 192.168.70.0 255.255.255.0
 default-router 192.168.70.1
 dns-server 8.8.8.8
```

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5. Switch Configuration

- **Trunks:** Carry multiple VLANs.
- **Access Ports:** Assigned to VLANs.

Example:

```
interface g1/0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70

interface fa0/2
 switchport mode access
 switchport access vlan 10
```

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6. Wireless (WRT300N)

- Connected to **VLAN 60** (Wi-Fi).
- DHCP disabled on WRT300N (router handles IPs).
- SSID configured for open/WPA2 access.

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7. ACL Firewall Policies

ACLs enforce security and segmentation:

- **Students (30,50) & Wi-Fi (60):**
  - Allowed → DMZ HTTP/HTTPS & DNS.
  - Denied → Admin (10) & IT (20).
- **DMZ (70):**
  - Denied → Internal VLANs.
  - Allowed → Outbound access.
- **Admin/IT (10,20):** Trusted VLANs with higher privileges.

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8. Switch Security

Layer 2 protections:

- **Port Security** (limit MACs per port).
- **DHCP Snooping** (blocks rogue servers).
- **Dynamic ARP Inspection (DAI)** (prevents ARP spoofing).
- **STP Protection** (BPDU Guard & PortFast).

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9. Device Hardening

Administrative access secured with passwords and encryption:

- **Enable Secret (Privileged Mode):** `SwitchEnable#123`
- **Console Password:** `class123`
- **VTY Password:** `class123`
- **Router Passphrase (simulated key):** `Student@123`

Additional measures:

- `service password-encryption` applied.
- Login banners warn unauthorized users.

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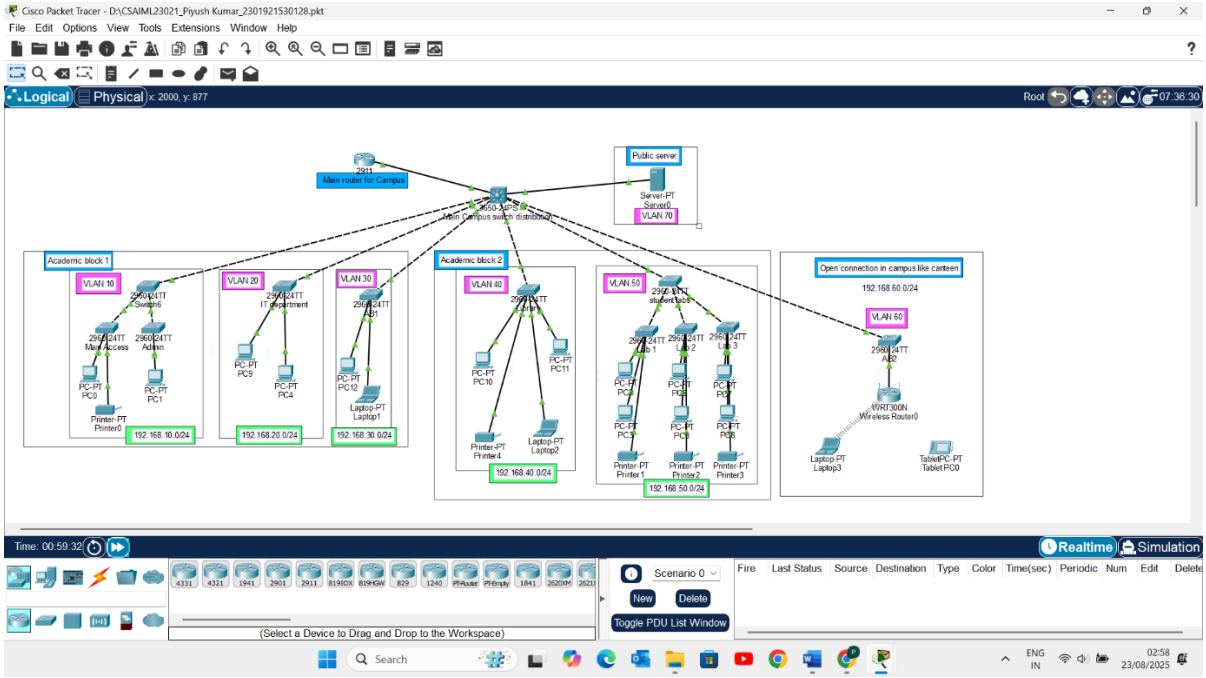
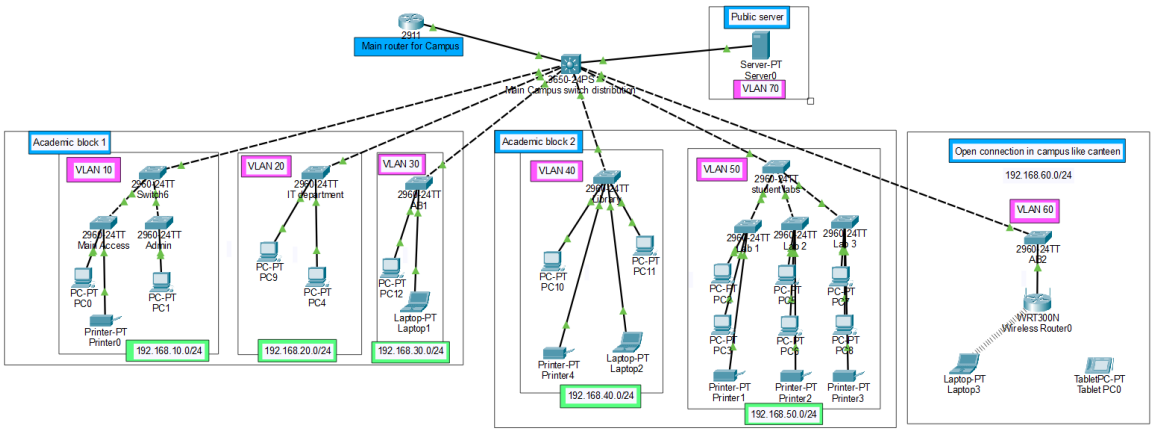
10. Verification & Testing

- **DHCP:** End devices get correct IPs.
- **ACLs:** Confirmed via ping/web tests.
- **Wi-Fi:** Clients on VLAN 60 receive IPs.
- **DMZ:** Accessible only for permitted services.
- **Commands used:**
  - `show vlan brief`
  - `show ip dhcp binding`
  - `show access-lists`

ScreenShot:


Cisco Packet Tracer - D:\CSAIML23021_Piyush Kumar_2301921530128.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
Public server
Main router for Campus
Server-PT
Server0
VLAN 70
Academic block 1
VLAN 10
VLAN 20
VLAN 30
192.168.10.0/24
192.168.20.0/24
192.168.30.0/24
Academic block 2
VLAN 40
VLAN 50
192.168.40.0/24
192.168.50.0/24
Open connection in campus like canteen
192.168.60.0/24
VLAN 60
Time: 00:59:32
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
(Select a Device to Drag and Drop to the Workspace)
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Search
ENG
IN
02:58
23/08/2025


Public server
Main router for Campus
Server-PT
Server0
VLAN 70
Academic block 1
VLAN 10
VLAN 20
VLAN 30
192.168.10.0/24
192.168.20.0/24
192.168.30.0/24
Academic block 2
VLAN 40
VLAN 50
192.168.40.0/24
192.168.50.0/24
Open connection in campus like canteen
192.168.60.0/24
VLAN 60